

Fentanyl-Piramal

Product Description
Clear, colourless solution. Free from visible foreign matter.

DOSAGE FORMS AND STRENGTHS

Fentanyl is a sterile, preservative-free, isotonic aqueous solution for intrave-nous use. Each ml contains 50 mcg fentanyl (*as fentanyl citrate*).
For excipients, *see List of Excipients*.

CLINICAL INFORMATION

- Indications**
FENTANYL-PIRAMAL is indicated:
- for use as an opioid analgesic supplement in general or regional anesthe-sia.
 - for administration with a neuroleptic as an anesthetic premedication, for the induction of anesthesia, and as an adjunct in the maintenance of general and regional anesthesia.
 - for use as an anesthetic agent with oxygen in selected high-risk patients undergoing major surgery.

Dosage and Administration
The dosage of FENTANYL-PIRAMAL should be individualized according to age, body weight, physical status, underlying pathological condition, use of other drugs, and type of surgery and anesthesia.

To avoid bradycardia, it is recommended to administer a small intravenous dose of an anti- cholinergic just before anesthetic induction.

Use as an analgesic supplement to general anesthesia

Low dose: 2 mcg/kg.
FENTANYL-PIRAMAL in small doses is most useful for minor, surgery.

Moderate dose: 2-20 mcg/kg.
Where surgery becomes more complicated, a larger dose will be required. The duration of activity is dependent on dosage.

High dose: 20-50 mcg/kg.
During major surgical procedures, in which surgery is longer, and during which the stress response would be detrimental to the well-being of the pa-tient, doses of 20-50 mcg/kg of FENTANYL- PIRAMAL with nitrous oxide/oxy-gen have been shown to have an attenuating effect. When doses in this range have been used during surgery, post-operative ventilation and observation are essential in view of the possibility of extended post-operative respiratory depression.

Supplemental doses of 25-250 mcg (0.5-5 mL) should be tailored to the needs of the patient and to the anticipated time until completion of the operation.

Use as an anesthetic agent

When attenuation of the response to surgical stress is especially important, doses of 50-100 mcg/kg may be administered with oxygen and a muscle re-laxant. This technique provides anesthesia without necessitating the use of additional anesthetic agents. In certain cases, doses up to 150 mcg/kg may be required to produce this anesthetic effect. FENTANYL-PIRAMAL has been used in this fashion for open heart surgery and certain other major surgical procedures in patients for whom protection of the myocardium from excess oxygen demand is particularly indicated.

Special Populations
Pediatrics
For induction and maintenance in children aged 2-12 years, a dose of 2-3 mcg/kg is recommended.

Elderly and debilitated patients
As with other opioids, the initial dose should be reduced in the elderly (> 65 years of age) and in debilitated patients. The effect of the initial dose should be taken into account in determining supplemental doses.

Obese patients
In obese patients there is a risk of overdosing if the dose is calculated based on body weight. Obese patients should be dosed based on estimated lean body mass rather than on body weight only.

Renal impairment
In patients with renal impairment reduced dosing of FENTANYL-PIRAMAL should be considered and these patients should be observed carefully for signs of fentanyl toxicity (*see Pharmacokinetic Properties*).

Contraindications
Known intolerance to any of its components or to other opioids.

Warnings and Precautions
Respiratory depression
As with all potent opioids, respiratory depression is dose related and can be

reversed by a specific opioid antagonist, but additional doses may be neces-sary because the respiratory depression may last longer than the duration of action of the opioid antagonist. Profound analgesia is accompanied by marked respiratory depression, which can persist or recur in the postopera-tive period. Therefore, patients should remain under appropriate surveillance. Resuscitation equipment and opioid antagonists should be readily available. Hyperventilation during anesthesia may alter the patient's responses to CO2, thus affecting respiration postoperatively.

Muscle rigidity
Induction of muscle rigidity, which may also involve the thoracic muscles, can occur, but can be avoided by the following measures: slow IV injection (ordi-narily sufficient for lower doses), premedication with benzodiazepines and the use of muscle relaxants.

Cardiac disease
Bradycardia, and possibly cardiac arrest, can occur if the patient has received an insufficient amount of anticholinergic, or when FENTANYL-PIRAMAL is combined with non-vagolytic muscle relaxants. Bradycardia can be treated with atropine.

Opioids may induce hypotension, especially in hypovolemic patients. Appro-priate measures to maintain a stable arterial pressure should be taken.

Special dosing conditions
The use of rapid bolus injections of opioids should be avoided in patients with compromised intracerebral compliance; in such patients the transient decrease in the mean arterial pressure has occasionally been accompanied by a short-lasting reduction of the cerebral perfusion pressure.

Patients on chronic opioid therapy or with a history of opioid abuse may require higher doses.

It is recommended to reduce the dosage in the elderly and in debilitated patients. Opioids should be titrated with caution in patients with any of the following conditions: uncontrolled hypothyroidism, pulmonary disease, decreased respiratory reserve, alcoholism, or impaired hepatic or renal function. Such patients also require prolonged post-operative monitoring.

Risks from concomitant use with CNS depressants especially benzodi-azepines or related drugs
The risk of profound sedation, respiratory depression, coma, and death may

be increased with the concomitant use of FENTANYL-PIRAMAL and CNS depressants, especially benzodiazepines or related drugs in spontaneous breathing patients. Observational studies have demonstrated that concomi-tant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone.

If the decision is made to newly prescribe a benzodiazepine or related drug and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine or related drug in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine or related drug than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine or related drug, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when FENTANYL-PIRAMAL is used with benzodi-azepines. Advice patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (*see Interactions*).

Interaction with neuroleptics
If FENTANYL-PIRAMAL is administered with a neuroleptic, the user should be familiar with the special properties of each drug, particularly the differ-ence in duration of action. When such a combination is used, there is a higher incidence of hypotension. Neuroleptics can induce extrapyramidal symptoms that can be controlled with anti-Parkinson agents.

Serotonin syndrome with concomitant use of serotonergic drugs
Caution is advised when FENTANYL-PIRAMAL is coadministered with drugs that affect the serotonergic neurotransmitter systems.

The development of a potentially life-threatening serotonin syndrome may occur with the concomitant use of serotonergic drugs such as Selective Serotonin Re-uptake Inhibitors (SSRIs) and Serotonin Norepinephrine Reuptake Inhibitors (SNRIs), and with drugs which impair metabolism of serotonin (including Monoamine Oxidase Inhibitors [MAOIs]). This may occur within the recommended dose.

Serotonin syndrome symptoms may include mental status changes (e.g., agitation, hallucinations, coma), autonomic instability (e.g., tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g., hyperreflexia, incoordination, rigidity), and/or gastrointestinal symptoms (e.g., nausea, vomiting, diarrhoea) and can be fatal (*see Interactions with Other Medicaments*).

The onset of symptoms may occur within minutes, hours or days of concomitant use. Discontinue FENTANYL-PIRAMAL if serotonin syndrome is suspected.

Drug dependence, tolerance and potential for abuse
Prolonged use of this product may lead to drug dependence (addiction), even therapeutic doses. The risks are increased in individuals with current or past history of substances misuse disorder (including alcohol misuse) or mental health disorder (e.g., major depression). Additional support and monitoring may be necessary when prescribing for patients at risk of opioid misuse. A comprehensive patient history should be taken to document concomitant medications, including over-the-counter medicines and medicines obtained on-line, and past and present medical and psychiatric conditions. Patients may find that treatment is less effective with chronic use and express a need to increase the dose to obtain the same level of pain control as initially ex-perienced. Patients may also supplement their treatment with additional pain relievers. These could be signs that the patient is developing tolerance. The risks of developing tolerance should be explained to the patient. Overuse or misuse may result in overdose and/or death. It is important that patients only use medicines that are prescribed for them at the dose they have been pre-scribed and do not give this medicine to anyone else. Patients should be closely monitored for signs of misuse, abuse, or addiction. The clinical need for analgesic treatment should be reviewed regularly.

Drug withdrawal syndrome
Prior to starting treatment with any opioids, a discussion should be held with patients to put in place a withdrawal strategy for ending treatment with fen-tanyl. Drug withdrawal syndrome may occur upon abrupt cessation of therapy or dose reduction. When a patient no longer requires therapy, it is advisable to taper the dose gradually to minimise symptoms of withdrawal. Tapering from a high dose may take weeks to months. The opioid drug withdrawal syndrome is characterised by some or all of the following: restlessness, lacrimation, rhinorrhoea, yawning, perspiration, chills, myalgia, mydriasis and palpitations. Other symptoms may also develop

including irritability, agitation, anxiety, hyperkinesia, tremor, weakness, in-somnia, anorexia, abdominal cramps, nausea, vomiting, diarrhoea, increased blood pressure, increased respiratory rate or heart rate.

Neonatal withdrawal syndrome
If women take opioids chronically during pregnancy, there is a risk that their newborn infants will experience neonatal withdrawal syndrome (*see Pregnancy*).
Opioid induced hyperalgesia
Opioid induced hyperalgesia (OIH) is a paradoxical response to an opioid, particularly at high doses or with chronic use, in which there is an increase in pain perception despite stable or increased opioid exposure. It differs from tolerance, in which higher opioid doses are required to achieve the same analgesic effect or treat recurring pain. OIH may manifest as increased levels of pain, more generalized pain (i.e., less focal), or pain from ordinary (i.e. non-painful) stimuli (allodynia) with no evidence of disease progression. When OIH is suspected, the dose of opioids should be reduced or tapered off, if possible.

Tolerance and Opioid use disorder (abuse and dependence)
Tolerance, physical dependence, and psychological dependence may develop upon repeated administration of opioids.

Repeated use of opioids may lead to Opioid use disorder (OUD). Abuse or intentional misuse of opioids may result in overdose and/or death. The risk of developing OUD is increased in patients with a personal or a family history (parents or siblings) of substance use disorders (Including alcohol use disorder), in current tobacco users or in patients with a personal history of other mental health disorders (e.g. major depression, anxiety and personality disorders).

Interactions
Effect of Other Drugs on FENTANYL-PIRAMAL
Central Nervous System (CNS) depressants
Drugs such as barbiturates, benzodiazepines, neuroleptics, general anes-thetics and other, non- selective CNS depressants (e.g. alcohol) may potenti-ate the respiratory depression of opioids. When patients have received such drugs, the dose of FENTANYL-PIRAMAL required may be less than usual. Concomitant use with FENTANYL-PIRAMAL in spontaneously breathing patients may increase the risk of respiratory depression, profound sedation, coma, and death.

Benzodiazepines
The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at µ-receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid- related respiratory depression exists.

Limit dosage and duration of concomitant use of benzodiazepines and opi-oids, and follow patients closely for respiratory depression and sedation.

Gabapentinoids (gabapentin and pregabalin)
Drugs such as barbiturates, benzodiazepines or related drugs, neuroleptics, general anaesthetics, gabapentinoids (gabapentin and pregabalin), and oth-er, non-selective CNS depressants (e.g., alcohol) may potentiate the respira-tory depression of opioids.

Cytochrome P450 3A4(CYP3A4) inhibitors
FENTANYL-PIRAMAL, a high clearance drug, is rapidly and extensively me-tabolized mainly by CYP3A4. When FENTANYL-PIRAMAL is used, the con-comitant use of a CYP3A4 inhibitor may result in a decrease in fentanyl clear-ance. With single-dose FENTANYL-PIRAMAL administration, the period of risk for respiratory depression may be prolonged, which may require special patient care and longer observation. With multiple-dose FENTANYL-PIRA-MAL administration, the risk for acute and/or delayed respiratory depression may be increased, and a dose reduction of FENTANYL-PIRAMAL may be required to avoid accumulation of fentanyl. Oral ritonavir (a potent CYP3A4 inhibitor) reduced the clearance of a single intravenous FENTANYL-PIRAMAL dose by two thirds, although peak plasma concentrations of fentanyl were not affected. However, itraconazole (another potent CYP3A4 inhibitor) at 200 mg/ day given orally for 4 days had no significant effect on the pharmacokinetics of a single intravenous FENTANYLPIRAMAL dose. Co-administration of other potent or less potent CYP3A4 inhibitors, such as voriconazole or fluconazole, and FENTANYL-PIRAMAL may also result in an increased and/or prolonged exposure to fentanyl.

Monoamine oxidase inhibitors (MAOIs)
It is usually recommended to discontinue MAOIs 2 weeks prior to any surgi-cal or anesthetic procedure. However, several reports describe the unevent-ful use of FENTANYL-PIRAMAL during surgical or anesthetic procedures in patients on MAOIs.

Serotonergic drugs
The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue FENTANYL-PIRAMAL if serotonin syndrome is suspected. Examples of serotonergic drugs are selective se-

rotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mir-tazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (*see Warnings and Precautions*).

Effect of FENTANYL-PIRAMAL on Other Drugs
Following the administration of FENTANYL-PIRAMAL, the dose of other CNS-depressant drugs should be reduced. This is particularly important after surgery, because profound analgesia is accompanied by marked respiratory depression, which can persist or recur in the postoperative period. Administra-tion of a CNS depressant, such as a benzodiazepine, during this period may disproportionately increase the risk for respiratory depression.

The total plasma clearance and volume of distribution of etomidate is de-creased by a factor of 2 to 3 without a change in half-life when administered with fentanyl. Simultaneous administration of FENTANYL-PIRAMAL and in-travenous midazolam results in an increase in the terminal plasma half-life and a reduction in the plasma clearance of midazolam. When these drugs are co- administered with FENTANYL-PIRAMAL their dose may need to be reduced.

Pregnancy, Breast-feeding and Fertility
Pregnancy
There are no adequate data from the use of FENTANYL-PIRAMAL in preg-nant women. Fentanyl can cross the placenta in early pregnancy. Studies in animals have shown some reproductive toxicity (*see Non-clinical Information*). The potential risk for humans is unknown.

Chronic use of an opioid during pregnancy may cause drug dependence in the neonate, leading to neonatal withdrawal syndrome.

Administration during childbirth (including cesarean section) is not recom-mended because fentanyl crosses the placenta and may suppress sponta-neous respiration in the newborn period. If FENTANYL-PIRAMAL is admin-istered, assisted ventilation equipment must be immediately available for the mother and infant if required. An opioid antagonist for the child must always be available.

Breast-feeding
Fentanyl is excreted into human milk. Therefore, breast-feeding or use of ex-pressed breast milk is not recommended for 24 hours following the admin-istration of this drug. The risk/benefit of breastfeeding following FENTANYL-PIRAMAL administration should be considered.

Fertility
There are no clinical data on the effects of fentanyl on male or female fertility. In animal studies, some tests on rats showed reduced female fertility at mater-nal toxic doses (*see Non-clinical Information*).

Effects on Ability to Drive and Use Machines
Patients should only drive or operate a machine if sufficient time has elapsed (at least 24 hours) after the administration of FENTANYL-PIRAMAL.

Adverse Reactions
Throughout this section, adverse reactions are presented. Adverse reactions are adverse events that were considered to be reasonably associated with the use of fentanyl citrate based on the comprehensive assessment of the available adverse event information. A causal relationship with fentanyl citrate cannot be reliably established in individual cases. Further, because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in clinical practice.

Clinical trial data
The safety of FENTANYL-PIRAMAL was evaluated in 376 subjects who partic-ipated in 20 clinical trials evaluating FENTANYL-PIRAMAL used as an an-esthetic. These subjects took at least one dose of FENTANYL-PIRAMAL and provided safety data. Adverse reactions, as identified by the investigator, re-ported for ≥1% of FENTANYL-PIRAMAL treated subjects in these studies are shown in Table 1.

Table 1. Adverse Reactions Reported by ≥ 1% of FENTANYL-PIRA-MAL-treated Subjects in 20 Clinical Trials of FENTANYL-PI-RAMAL	
System/Organ Class Adverse Reaction	FENTANYL (n=376)%
Nervous System Disorders	
Sedation	5.3
Dizziness	3.7
Dyskinesia	3.2
Eye Disorders	
Visual disturbance	1.9

Cardiac Disorders	
Bradycardia	6.1
Tachycardia	4.0
Arrhythmia	2.9
Vascular Disorders	
Hypotension	8.8
Hypertension	8.8
Vein pain	2.9
Respiratory, Thoracic and Mediastinal Disorders	
Apnea	3.5
Bronchospasm	1.3
Laryngospasm	1.3
Gastrointestinal Disorders	
Nausea	26.1
Vomiting	18.6
Skin and Subcutaneous Tissue Disorders	
Dermatitis allergic	1.3
Musculoskeletal and Connective Tissue Disorders	
Muscle rigidity (which may also involve the thoracic muscles)	10.4
Injury, Poisoning and Procedural Complications	
Confusion postoperative	1.9
Anesthetic complication neurological	1.1

Additional adverse reactions that occurred in <1% of FENTANYL-treated subjects in the 20 clinical trials are listed below in Table 2.

Table 2. Adverse Reactions Reported by < 1% of FENTANYL- PIRA-MAL treated Subjects in 20 Clinical Trials of FENTANYL- PIRAMAL	
System/Organ Class	
Adverse Reaction	
Psychiatric Disorders	
Euphoric mood	
Nervous System Disorders	
Headache	
Vascular Disorders	
Blood pressure fluctuation	
Phlebitis	
Respiratory, Thoracic and Mediastinal Disorders	
Hiccups	
Hyperventilation	
General Disorders and Administration Site Conditions	
Chills	
Hypothermia	
Injury, Poisoning and Procedural Complications	
Agitation postoperative	
Procedural complication	
Airway complication of anesthesia	

Postmarketing data
Adverse reactions first identified during postmarketing experience with FENTANYL are included in Table 3. In the table, the frequencies are provided according to the following convention:

Very common	≥ 1/10
Common	≥ 1/100 and < 1/10
Uncommon	≥ 1/1,000 and < 1/100
Rare	≥ 1/10,000 and < 1/1,000
Very rare	< 1/10,000, including isolated reports

In Table 3, the adverse reactions are presented by frequency category based on incidence in clinical trials or epidemiology studies, when known. The frequency category “not known” is used for adverse reactions for which no valid estimate of the incidence rate can be derived from clinical trials.

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Table 3: Adverse Reactions Identified During Postmarketing Experience with FENTANYL-PIRAMAL by Frequency Category Estimated from Clinical Trials or Epidemiologic Studies

Immune System Disorders	
<i>Not Known</i>	Hypersensitivity (such as anaphylactic shock, anaphylactic reaction, urticaria)
Psychiatric Disorders	
<i>Not Known</i>	Delirium, Drug dependence (<i>see Warnings and Precautions</i>)
Nervous System Disorders	
<i>Not Known</i>	Convulsions, Loss of consciousness, Myoclonus
Cardiac Disorders	
<i>Not Known</i>	Cardiac arrest (<i>see Warnings and Precautions</i>)
Respiratory, Thoracic and Mediastinal Disorders	
<i>Not Known</i>	Respiratory depression (<i>see Warnings and Precautions</i>)
Skin and Subcutaneous Tissue Disorders	
<i>Not Known</i>	Pruritus
General Disorders and Administration Site Conditions	
<i>Not Known</i>	Drug withdrawal syndrome (<i>see Warnings and Precautions</i>)

When a neuroleptic is used with FENTANYL-PIRAMAL, the following adverse reactions may be observed: chills and/or shivering; restlessness, post-operative hallucinatory episodes; and extrapyramidal symptoms (*see Warnings and Precautions*).

Overdose
Symptoms and signs
An overdosage of FENTANYL-PIRAMAL manifests itself as an extension of its pharmacologic actions. Respiratory depression which can vary in severity from bradypnea to apnea may occur.

Treatment
In the presence of hypoventilation or apnea, oxygen should be administered and respiration should be assisted or controlled as indicated. A specific opioid antagonist should be used as indicated to control respiratory depression. This does not preclude the use of more immediate countermeasures. The respiratory depression may last longer than the effect of the antagonist; additional doses of the latter may therefore be required.

If depressed respiration is associated with muscular rigidity, an intravenous neuromuscular blocking agent might be required to facilitate assisted or controlled respiration.

The patient should be carefully observed; body warmth and adequate fluid intake should be maintained. If hypotension is severe or if it persists, the possibility of hypovolemia should be considered, and if present, should be controlled with appropriate parenteral fluid administration.

PHARMACOLOGICAL PROPERTIES
Pharmacodynamic Properties
Pharmacotherapeutic group: anesthetics general, opioid anesthetics. ATC Code N01AH01

Mechanism of action
Fentanyl is a potent, opioid analgesic.

Pharmacodynamic effects
Fentanyl is an opioid analgesic interacting predominantly with the μ -opioid receptor. Fentanyl can be used as an analgesic supplement to general anesthesia or as the sole anesthetic. Fentanyl preserves cardiac stability and obtunds stress-related hormonal changes at higher doses. A dose of 100 mcg (2.0 mL) is approximately equivalent in analgesic activity to 10 mg of morphine. The onset of action is rapid. However, the maximum analgesic and respiratory depressant effect may not be noted for several minutes. The usual duration of action of the analgesic effect is approximately 30 minutes after a single IV dose of up to 100 mcg. Depth of analgesia is dose- related and can be adjusted to the pain level of the surgical procedure.

Like other opioid analgesics, FENTANYL-PIRAMAL, depending upon the dose and speed of administration, can cause muscle rigidity, as well as euphoria, miosis and bradycardia.

Histamine assays and skin-wheel testing have indicated that clinically significant histamine release is rare with FENTANYL-PIRAMAL.

All actions of FENTANYL-PIRAMAL are reversed by a specific opioid antagonist.
Pharmacokinetic Properties
Distribution
After intravenous injection, fentanyl plasma concentrations fall rapidly, with sequential distribution half-lives of about 1 minute and 18 minutes and a terminal

elimination half-life of 475 minutes. Fentanyl has a Vc (volume of distribution of the central compartment) of 13 L, and a total Vdss (distribution volume at steady-state) of 339 L. The plasma-protein binding of Fentanyl is about 84 %.

Metabolism
Fentanyl is rapidly metabolized, mainly in the liver by CYP3A4. The major metabolite is norfentanyl. Fentanyl clearance is 574 mL/min.

Excretion
Approximately 75 % of the administered dose is excreted in the urine within 24 hours and only 10 % of the dose eliminated in urine is present as unchanged drug.

Special Populations
Pediatrics
The plasma protein binding of fentanyl in newborns is approximately 62 % which is lower than in adults. The clearance and the volume of distribution are higher in infants and children. This may result in an increased dose requirement for FENTANYL-PIRAMAL.

Renal impairment
Data obtained from a study administering IV fentanyl in patients undergoing renal transplantation suggest that the clearance of fentanyl may be reduced in this patient population. If patients with renal impairment receive FENTANYL-PIRAMAL, they should be observed carefully for signs of fentanyl toxicity and the dose reduced if necessary (*see Dosage and Administration*).

Adult patients with burns
An increase in clearance up to 44 % together with a larger volume of distribution results in lower fentanyl plasma concentrations. This may require an increased dose of FENTANYL-PIRAMAL.

Obese patients
An increase in clearance of fentanyl is observed with increased body weight. In patients with a BMI>30, clearance of fentanyl increases by approximately 10% per 10 kg increase of the fat free mass (lean body mass).

NON-CLINICAL INFORMATION
Fentanyl has a broad safety-margin. In rats the ratio LD₅₀/ED₅₀ for the lowest level of analgesia is 281.8, as compared with 69.5 and 4.8 for morphine and pethidine respectively.

Carcinogenicity and Mutagenicity
In vitro fentanyl showed, like other opioid analgesics, mutagenic effects in a mammalian cell culture assay, only at cytotoxic concentrations and along with metabolic activation. Fentanyl showed no evidence of mutagenicity when tested in *in vivo* rodent studies and bacterial assays. In a two-year carcinogenicity study conducted in rats, fentanyl was not associated with an increased incidence of tumors at subcutaneous doses up to 33 mcg/kg/day in males or 100 mcg/kg/day in females, which were the maximum tolerated doses for males and females.

Reproductive Toxicology
Fertility
Some tests on female rats showed reduced fertility as well as embryo mortality. These findings were related to maternal toxicity and not a direct effect of the drug on the developing embryo.

There was no evidence of teratogenic effects.

PHARMACEUTICAL INFORMATION
List of Excipients
Sodium chloride
Water for injections

Incompatibilities
The injectable solution must not be mixed with other products.

If desired, FENTANYL-PIRAMAL may be mixed with sodium chloride or glucose intravenous infusions. Such dilutions are compatible with plastic infusion sets. These should be used within 24 hours of preparation.

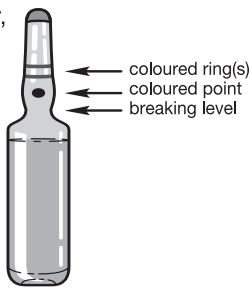
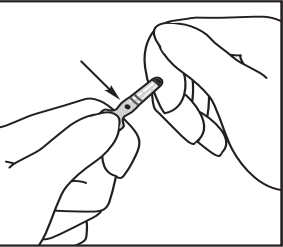
Shelf Life
3 years

Storage Conditions
Protect from light.
Store below 30°C.
Keep out of sight and reach of children.

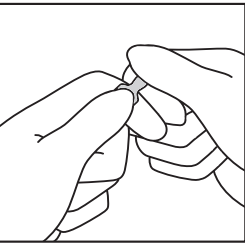
Nature and Contents of Container
FENTANYL-PIRAMAL is supplied in 2 mL x 10 ampoules.

Instructions for Use and Handling
Wear gloves while opening ampule.

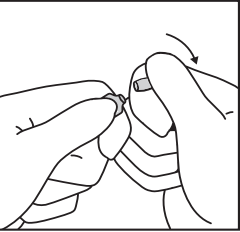
Maintain the ampule between thumb and index finger, leaving the tip of the ampule free.



With the other hand, hold the tip of ampule, putting the index finger against the neck of ampule and the thumb on the colored point in parallel to the identification colored ring(s).



Keeping the thumb on the point, sharply break the tip of ampule while holding firmly the other part of the ampule in the hand.



Accidental dermal exposure should be treated by rinsing the affected area with water. Avoid use of soap, alcohol, and other cleaning materials that may cause chemical or physical abrasions to the skin.

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DATE OF REVISION OF THE TEXT
14/07/2022

Parameters	Remarks	Parameters	Remarks
Product name/Generic name	Fentanyl 50 x 2 ml PIL Malaysia	Market/Country	Malaysia
Strength	NA	Barcode (PZN/CIP/EAN/KD/SKUL/VNR)	NA
Component	Pack Insert	Material code	38885330
Dimensions	696 x 332 mm	Superseded Material code	NA
Specification	NA	Artwork number	NA
Font size & type	Helvetica World	MAH	NA
Pantone number	As shown	GTIN	NA
Cut margins/Peel off Margins	NA	Special instructions	NA
Unvarnished zone	NA	Anticounterfiet features	NA
Mfg site	NA	Other observation	NA
Reason for revision:	New Artwork	Pharma Code	439
Date: 14/07/2022			

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